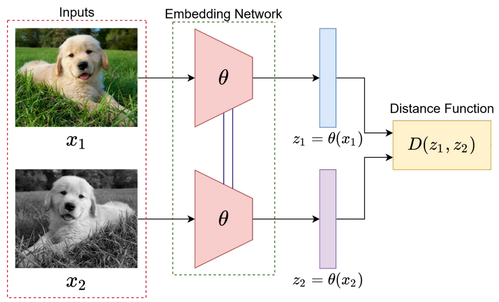
Self-supervised learning for robot grasping

Kirill Gelvan & Denis Grachev

Team 02, TUM

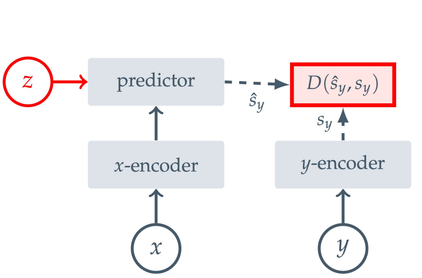
{kirill.gelvan, denis.grachev}@tum.de

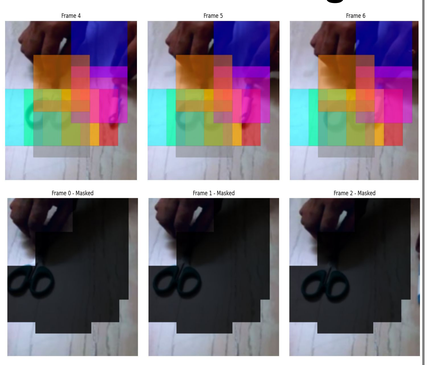
Can Self-Supervised Learning (SSL) help with Grasping?



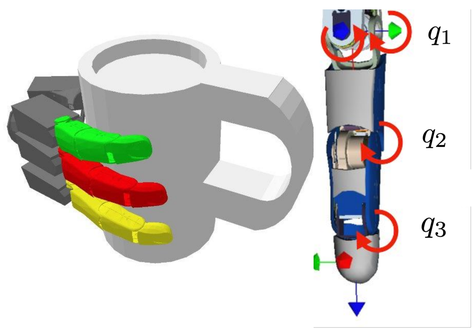
- **3D objects:**
  - ➢ A lot (ShapeNet >50k)
- **Grasps:**
  - ➢ Hundreds
- **Goal:**
  - ➢ Improve grasping quality

Joint Embedding Predictive Architecture & Masking





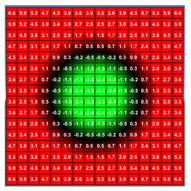
Grasping



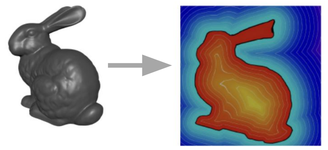
- (12+1)-d regression task
  - ➢ Joints & Accuracy
- 15K objects with 24 grasps each
- Qualitative and quantitative evaluation
  - ➢ MAE & Angle error
  - ➢ Grasping pictures

Signed Distance Fields (SDF) and Point Clouds

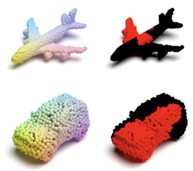
SDF

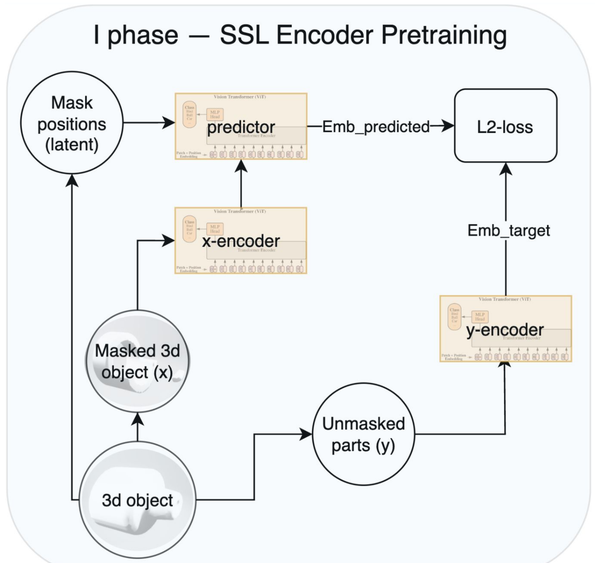


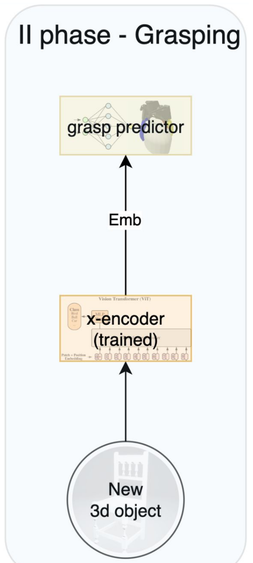
3D Mesh to SDF



Point Clouds

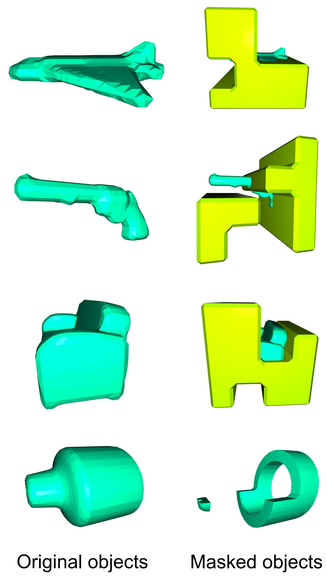






Masking

- Remake masks from video to 3D from scratch
- Try different masking techniques
  - ➢ 8 intersecting masks is the best



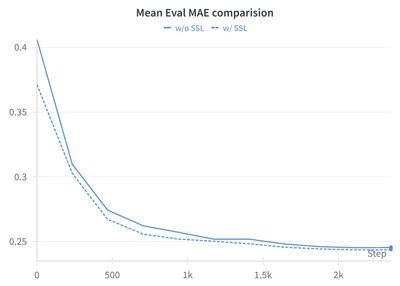
Original objects Masked objects

Training SSL

- Vision Transformer from scratch
- emb_size=192, layers=12, ~5M parameters
- Objects are 32x32x32 pixels

Training Grasping

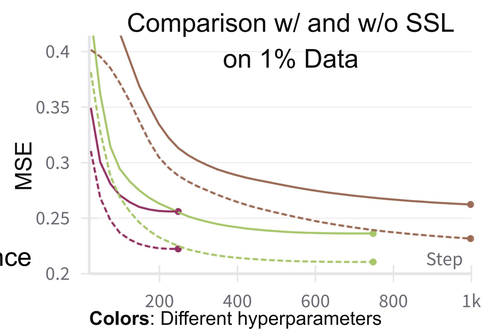
- Hand position is fixed
- Predict Score & Joint Angles
- ViT + head ⟶ regression
  - ➢ Difference is negligible after training



Results

- ViT vs ResNet
  - ➢ No significant difference
- Frozen vs Unfrozen
  - ➢ SSL is not good enough
- Available Grasping data (1/10/100%)
  - ➢ On 1% of the dataset (~3000 grasps)

    SSL **outperforms** & preserves the difference



Colors: Different hyperparameters

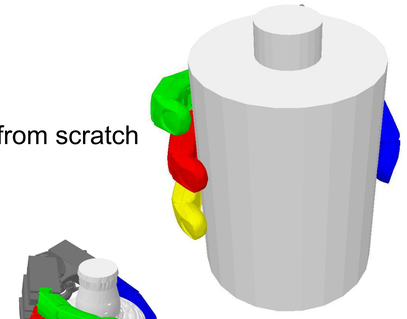
Lines: Dotted (w/ SSL) vs Straight (w/o SSL)

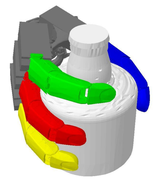
Table 1: Comparison of Approaches with Varying Data Amounts

Data Amount	MSE w/o SSL	MSE w/ SSL
100% data	0.1176	0.1175
10% data	0.1632 ± 0.0036	0.1602 ± 0.0043
1% data	0.2353 ± 0.0122	$\mathbf{0.2058 \pm 0.0078}$

Conclusions

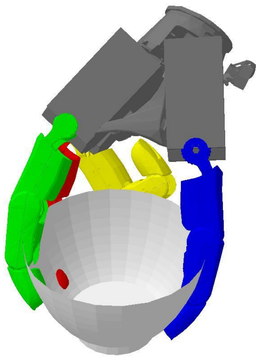
- Trained over 400 models
- Developed SSL and grasping from scratch
- SSL is beneficial:
  - ➢ **For small datasets**
  - ➢ At the start of the training
  - ➢ For training acceleration





Future Work

- Conduct more experiments on ViT, embedding sizes and hyperparameter search
- Add more datasets for SSL
- Test SSL on other 3D tasks



Video samples

WandB report

DLR

*all illustrated grasps are predicted by trained models on validation examples